

Sumit Sharma

Software Engineer (Full-Stack) | B.Tech – Computer Science & Engineering (Data Science)

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PROFESSIONAL SUMMARY

Full-stack Software Engineer and B.Tech CSE student who ships production web applications with React 18/19, TypeScript, Vite, Tailwind CSS, FastAPI and PostgreSQL. Built and deployed the live ITM Gwalior website plus two full-stack products, TestHub and VNotes. Strong in REST API design, authentication (JWT, OAuth 2.0, RBAC), frontend performance and SEO, Docker and cloud deployment (Vercel, Google Cloud Run, Render, CI/CD).

EXPERIENCE

Web Development Intern (Frontend Lead)

Jan 2025 – Present

ITM Gwalior

Gwalior, India

- Own the frontend of the institute's production website and CMS (itmgoi.in, see Projects): React 19 architecture, performance, SEO and admin tooling, shipped continuously on Vercel.
- Contributed to the backend: FastAPI, PostgreSQL and Redis services with 340+ REST endpoints, JWT access/refresh rotation, Argon2 hashing, rate limiting and Cloud Build CI/CD.

Data Analytics Intern

May 2025 – Jun 2025

Navodita Infotech

Remote

- Performed data cleaning, preprocessing and exploratory data analysis on client datasets in Python (Pandas, NumPy) and delivered visualizations and reports that informed data-driven business decisions.

PROJECTS

ITM Gwalior Website & CMS – Production | React 19, Vite 7, Tailwind CSS, TanStack Query, FastAPI, Vercel

Live Site

- Built and shipped the official institute website and admin CMS: 164 routes, 64 public pages, 17 admin modules and a dynamic-page route so admin-created URLs go live without a redeploy; dark mode, Framer Motion, GA4 and Vercel Analytics.
- Optimized performance with 127 lazy-loaded routes, manual vendor/admin chunking (97 chunks so admin code never ships to visitors), viewport-deferred home sections, lazy images and immutable asset caching headers.
- Engineered SEO: zero-dependency prerender script emitting static HTML with per-route meta for 95 routes, dynamic sitemap generator (static, CMS and blog URLs) and a custom head manager for OpenGraph, Twitter cards and JSON-LD schema.
- Built the admin CMS UI: draft/publish page editor with 6 block types, live in-place editing on the public site (contentEditable, debounced saves, cross-tab sync), scope-gated components over 60 RBAC scopes, Axios single-flight JWT refresh.

TestHub – Secure Online Exam Platform | React 18, Vite, FastAPI, SQLAlchemy, PostgreSQL, JWT

Live Demo

- Built a full-stack proctored exam platform: 39 REST endpoints across 4 routers, 7 PostgreSQL tables (async SQLAlchemy, asyncpg), 17 frontend routes; three login flows (password, hashed email OTP with Brevo/Resend/SMTP fallback, Google OAuth 2.0) and three-scope JWTs (admin, user, attempt).
- Engineered an 8-signal anti-cheat hook (tab switch, fullscreen exit, DevTools debugger-trap and window-delta detection, clipboard and shortcut blocking) with server-side violation thresholds, server-authoritative deadlines, answer-leak-safe schemas and frozen shuffled question sets; bulk question import from JSONL, JSON, CSV and PDF; deployed on Vercel and Render.

VNotes – AI YouTube Notes Generator | React, TypeScript, Vite, Tailwind, TanStack Query, FastAPI

Live Demo

- Built a TypeScript React SPA with Google One Tap login exchanged for a JWT and protected routes; adaptive TanStack Query polling (3 s while jobs run, off when idle) drives a 6-step live progress stepper; Markdown + KaTeX rendering, Markdown download and print-to-PDF.
- Designed the FastAPI backend: transcript, Whisper speech-to-text and LLM note generation as background jobs with a 7-state status machine, tiered daily quotas (HTTP 429), Alembic migrations, Docker on Google Cloud Run.

Movie Recommendation Web App | Python, Flask, Jinja2, REST APIs, JavaScript, CSS

- Built a server-rendered Flask app integrating TMDB, OMDb and YouTube Data v3 APIs (search, recommendations, detail pages with trailers) with a shared HTTP session, 3-attempt exponential-backoff retries, error pages and a light/dark CSS theme system.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, SQL, C++, C, HTML5, CSS3

Frontend: React 18/19, Vite, Tailwind CSS, React Router, TanStack Query, Framer Motion, Code Splitting, SSG, SEO, Responsive Design

Backend: FastAPI, Flask, REST API Design, SQLAlchemy (async), Alembic, Pydantic, JWT, OAuth 2.0, RBAC, SSE, Caching

Databases: PostgreSQL, Redis, MySQL, Supabase, pgvector

DevOps & Tools: Docker, Vercel, Google Cloud Run, Cloud Build CI/CD, Render, Git/GitHub, Linux, pytest, ESLint, npm

AI Integration: LLM APIs (Groq, NVIDIA NIM, Gemini), RAG, LangChain, Hugging Face, Whisper

EDUCATION

Institute of Technology and Management, Gwalior

2023 – 2027

B.Tech – Computer Science & Engineering (Data Science) | CGPA: 7.50

Gwalior, India

PUBLICATIONS & CERTIFICATIONS

- Research Papers (IJAMRED, 2025):** Neuron Lifecycle in Deep Networks; Data-Driven Sustainability: Cognitive Decision Intelligence to Reduce Medical Waste in Pharmaceutical Logistics
- Research Chapter:** IIP International Publication (2026) | **Certification:** Oracle Fusion AI Agent Studio Foundations Associate